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My research interests are in probability and combinatorics, with focus on problems from network and data science, statistical physics, and pure combinatorics. My current work includes projects related to interacting particle systems, mixing times for Markov chains, random matrices, and symbolic dynamics.

Employment

- (2026-) Postdoctoral researcher, NYU Shanghai (Shanghai, China).
Faculty supervisor: [Alejandro Ramirez](#).
- (2023-2026) Postdoctoral researcher, Alfréd Rényi Institute of Mathematics (Budapest, Hungary).
Faculty supervisor: [Gábor Pete](#).
- (2020-2023) Postdoctoral researcher, University of British Columbia (Vancouver, Canada).
Faculty supervisors: [Omer Angel](#) and [Gordon Slade](#).

Education

- (2014-2020) Ph.D. in Mathematics, University of Washington.
Dissertation: *Random Combinatorial Processes*.
Ph.D. advisor: [Christopher Hoffman](#).
- (2010-2014) B.A. in Mathematics and Chinese Language, Dartmouth College.
Mathematics honors thesis: *Counting Clusters on a Grid*.

Papers

Published

1. *Shifts of Finite Type Obtained by Forbidding a Single Pattern*. Nishant Chandgotia, Brian Marcus, Jacob Richey, and Chengyu Wu. Discrete Contin. Dyn. Syst. 48: 538–576 (2026). doi: 10.3934/dcds.2025152. arXiv:2409.09024. 39 pages.
2. *Word Length, Bias and Bijections in Penney's Ante*. Matthew Drexel*, Xuanshan Peng*, and Jacob Richey. Electron. J. Combin. 32(4): #P4.57 (2025). doi: 10.37236/13609. arXiv:2409.19195. 23 pages. *Undergraduate coauthors at the time of the work.
3. *Diffusion-Limited Annihilating-Coalescing Systems*. Sungwon Ahn, Matthew Junge, Hanbaek Lyu, Lily Reeves, Jacob Richey, and David Sivakoff. Electron. J. Probab. 30: 1–20 (2025). doi: 10.1214/25-EJP1286. arXiv:2305.19333. 20 pages.

4. *Random Walks on Regular Trees Can Not Be Slowed Down*. Omer Angel, Jacob Richey, Yinon Spinka, and Amir Yehudayoff. Electron. J. Probab. 29: 1–15 (2024). doi: 10.1214/24-EJP1109. arXiv:2302.00760. 15 pages.
5. *Active Phase for Activated Random Walk on $\mathbb{Z}$* . Chris Hoffman, Jacob Richey, and Leonardo Rolla. Commun. Math. Phys. 399: 717–735 (2023). doi: 10.1007/s00220-022-04572-x. arXiv:2009.09491. 19 pages.
6. *Intersections of Random Sets*. Jacob Richey and Amites Sarkar. J. Appl. Probab. 59(1): 131–151 (2022). doi: 10.1017/jpr.2021.34. arXiv:2006.01323. 21 pages.
7. *Rumor Source Detection with Multiple Observations under Adaptive Diffusion Protocols*. Miklós Z. Rácz and Jacob Richey. IEEE Trans. Netw. Sci. Eng. 8(1): 2–12 (2021). doi: 10.1109/TNSE.2020.3022621. Extended version: arXiv:2006.11211. 11 pages.
8. *Activated Random Walk on a Cycle*. Riddhipratim Basu, Shirshendu Ganguly, Chris Hoffman, and Jacob Richey. Ann. Inst. H. Poincaré Probab. Statist. 55(3): 1258–1277 (2019). doi: 10.1214/18-AIHP918. arXiv:1709.09163. 20 pages.
9. *A Smooth Transition from Wishart to GOE*. Miklós Z. Rácz and Jacob Richey. J. Theoret. Probab. 32: 898–906 (2019). doi: 10.1007/s10959-018-0808-2. arXiv:1611.05838. 9 pages.

Accepted

10. *Explosivity in 1-d Activated Random Walk*. Nicolas Forien, Christopher Hoffman, Tobias Johnson, Josh Meisel, Jacob Richey, and Leonardo T. Rolla. To appear in Electron. Commun. Probab. arXiv:2601.03411 (2026). 7 pages.
11. *The Odometer in Subcritical Activated Random Walk*. Tobias Johnson and Jacob Richey. To appear in Electron. Commun. Probab. arXiv:2510.05514 (2025). 11 pages.

Submitted

12. *Letter Frequency in Shifts of Finite Type with One Forbidden Word*. Miklós Bóna, Balázs Maga, and Jacob Richey. Submitted to Combinatorial Theory. arXiv:2606.06655 (2026). 23 pages.
13. *Law of Large Numbers for Activated Random Walk on Villages*. Balázs Ráth, Jacob Richey, and Miklós Salánki. Submitted to Electron. Commun. Probab. arXiv:2605.06887 (2026). 12 pages.
14. *Local Density of Activated Random Walk on $\mathbb{Z}$* . Christopher Hoffman, Jacob Richey, and Hyojeong Son. Submitted to Proc. AMS. arXiv:2601.07816 (2026). 13 pages.

arXiv Preprints

15. *Active Phase for the Stochastic Sandpile on $\mathbb{Z}$* . Chris Hoffman, Yiping Hu, Jacob Richey, and Douglas Rizzolo. arXiv:2212.08293 (2022). 44 pages.

University Teaching

- (Spring 2025) Taught a graduate-level probability topics course at BME (Budapest University of Technology and Economics), titled *A Random Walk in a Random Park*. Topics: random walks, Bernoulli percolation, and first-passage percolation in $\mathbb{Z}^d$. Class size: 15. Students presented research papers for credit.
- (2020-2023) Lead instructor for courses in probability and calculus at the University of British Columbia. Class sizes ranged from 80 to 120. I was often in charge of one or two TAs or graders.
 - MATH 302, Introduction to Probability. Terms taught: Su23, W22, Su21.
 - MATH 102, Differential Calculus with Applications to Life Sciences. W21.
 - MATH 303, Introduction to Stochastic Processes. W20.
- (2016-2020) Lead instructor for undergraduate courses in advanced multivariable calculus and linear algebra at the University of Washington. Class sizes ranged from 30 to 50.
 - MATH 308, Matrix Algebra with Applications. Quarters taught: W20, Sp18.
 - MATH 324, Advanced Multivariable Calculus I. F19, Sp19, F18, W18, F17, Su17, Sp17, W17, Su16.
- (2014-2016) Teaching assistant for courses in precalculus and calculus at the University of Washington. Led homework sections, usually two groups of 50 students, and helped with exam grading.
 - MATH 120, Precalculus. Quarters taught: F16, Sp16, F15.
 - MATH 111, Algebra with Applications. W15.
 - MATH 112, Application of Calculus to Business and Economics. Sp15.
 - MATH 125, Calculus with Analytic Geometry II. Su15, F14.

Advising & Mentorship

- (Summer 2026) [NYC Discrete Math REU](#). Co-mentor for two groups of undergraduates in projects related to activated random walk, including ongoing research.
- (2025-26) Advised a master's student, Miklós Salánki, on his thesis project at the [Budapest University of Technology and Economics](#). Topic: the village model of activated random walk. The project led to the submitted paper *Law of Large Numbers for Activated Random Walk on Villages*.
- (2021-2023) [Undergraduate Research Opportunities \(REX\)](#) at UBC. Mentor for two year-long undergraduate research projects (three students each) in 2021–22 and 2022–23, culminating poster sessions. The Penney's ante project resulted in a paper published in the *Electronic Journal of Combinatorics* (2025). Topics: street-light percolation and coin bias in Penney's ante.

- (2017-2020) [Washington Experimental Mathematics Lab \(WXML\)](#). Co-led two undergraduate research projects, usually three or four students, over the course of a year or more, culminating in a poster presentation. Topics: statistics for random walks and randomness of the discrete logarithm.
- (Summer 2015) [University of Washington Inverse Problems REU](#). Gave a lecture at the REU colloquium and advised an undergraduate research project that resulted in an [REU research paper](#). Topic: counting clusters in Bernoulli percolation on lattices.

Other Teaching & Outreach

- (Spring 2024) Taught a two-month minicourse on topics in large deviations at the Rényi Institute for graduate students, postdocs, and faculty.
- (September 2022) Volunteer at STEM Britannia science event for youth, Vancouver, BC.
- (2016-2021) Ran the advanced course at UW Math Circle, an NSF-supported mathematics outreach program for advanced middle and high school students. Designed, tested, and taught new interactive lessons and activities in game theory, group theory, geometry, probability, and number theory.
- (Summer 2015) Junior staff at Hampshire College Summer Studies in Mathematics. Led problem sessions, wrote and graded problem sets, taught a minicourse, and gave a “prime time” lecture. Six days a week, eight hours a day for two months.
- (2012-2014) Private tutor for students in mathematics courses at Dartmouth College.

Organizing & Service

- (May 2025) Organizer, [Focused Workshop on Activated Random Walk at the Erdős Center](#). Led a week-long research workshop on activated random walk (11 participants, including 7 international visitors).
- (2022) Co-organizer, PIMS Probability Summer School. Coordinator for invited speakers.
- (2021-2023) Organizer, UBC probability seminar. Selected and coordinated speaker visits, including virtual talks, and maintained the [seminar website](#) and YouTube channel.
- Peer reviewer for *Proceedings of the AMS*, *Journal of the London Mathematical Society*, *Electronic Journal of Probability*, *Stochastic Models*, and *Information and Computation*.
- (2023-) Reviewer for Mathematical Reviews (10 reviews).

Invited Talks

- [Entropy and Statistics for Shifts of Finite Type](#)
 - West Coast Coin Flippers Union '26 (University of Delaware).

- Random Structures and Algorithms '25 (Vienna, Austria).
- University of Illinois Chicago discrete mathematics seminar, November '24.
- University of Chicago probability seminar, November '24.
- Los Angeles Probability Forum, October '24.
- Kutzem Research Seminar '24 (Rényi Institute, Budapest).
- Rutgers combinatorics seminar '23.
- PIMS dynamics seminar '23 (UBC, Vancouver, Canada).
- *Finding the Source of a Random Diffusion*
 - Cornell probability seminar '23.
 - CUNY probability seminar '23.
 - Dartmouth colloquium '21.
 - University of Victoria probability seminar '21.
- *Phase Transition for Activated Random Walk and the Stochastic Sandpile*
 - Universidad Nacional Autónoma de México, January '26
 - Indiana University Bloomington probability seminar, November '24.
 - Purdue probability seminar, November '24.
 - Ohio State University probability seminar, November '24.
 - University of Wisconsin–Madison probability seminar, October '24.
 - Kutzem Research Seminar '23 (Rényi Institute, Budapest).
 - University of Delaware probability seminar '23.
 - Cornell probability seminar '20.
 - Random Structures and Algorithms '23 (Carnegie Mellon, Pittsburgh).
 - CRM Workshop '22 (Montreal).
 - AMS Special Session on Stochastic Spatial Models, JMM '20 (Denver).
- *Rumor Source Detection with Multiple Observations under Adaptive Diffusion Protocols*
 - SIAM Workshop on Network Science '18 (Portland, OR).
 - DYNASNET Workshop '23 (Lednice, Czech Republic).
- *Phase Transition for Parking with Coalescence*
 - Northwest Probability Seminar '22.

Selected Conferences & Visits

- Attended and presented a talk
 - Random Structures and Algorithms, July 2025 (Vienna, Austria), and June 2023 (Pittsburgh, PA).
 - Northwest Probability Seminar, October 2022, Seattle, WA.
 - CRM Workshop on Interacting Particle Systems and Hydrodynamic Limits, March 2022, Montreal, Canada.
 - JMM 2020, Denver, CO.
- Attended
 - Saint-Flour Probability Summer School, July 2024, Saint-Flour, France.
 - Permutations and Probability (BIRS), September 2021, Banff, Canada.
 - AMS MRC on Stochastic Spatial Models, Summer 2019, Providence, RI.
 - Virginia Integrable Probability Summer School 2019, University of Virginia, Charlottesville, VA.
 - Visitor at NYU Shanghai for two weeks in May 2019. Sponsor: Leonardo Rolla. Shanghai, China.
 - Northwestern Probability Summer School, 2018, Northwestern University, Evanston, IL.
 - Recent Trends in Continuous and Discrete Probability, 2018, Georgia Tech, Atlanta, GA.

Awards

- Gerald B. Folland Fellowship (2019) — graduate student award for research excellence.

Other

- U.S. citizen.
- Languages: English; Mandarin Chinese (conversational); Spanish (intermediate).
- Technical skills: Python, Mathematica, MATLAB, $\text{\LaTeX}$.

Last updated: August 24, 2026